MAKE THE HUM VISIBLE

A free first investigation: a seven-night screening guide for recurring low-frequency hum.

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> **RELEASE STATUS: ANALYTICAL — NOT FIELD-VALIDATED.**
>
> No phone configuration has been tested. No field trial has been run. No nontechnical user has completed this guide. See Appendix C for exactly what is unverified and Appendix D for the open work.
>
> **Validated configuration: [PHONE MODEL / OS / APP — pending]**
>
> This is not a regulatory measurement procedure, and it is not legal, medical, or engineering advice.

What this guide does: it helps you record the pitch, timing, and recurrence of a hum you hear at night, using a phone you already own, for $0. It does not measure how loud the hum is. Tags like [A] and [S] show how well a claim is supported; they are explained under Document notes at the end.

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0. READ THIS FIRST — SHOULD YOU DOCUMENT IT AT ALL?

Sections 1 to 3 take **one evening**. They may help you sleep. They may also keep you from wasting money. Do them first.

Section 4 takes **seven nights**. You may not have that to give. Before you start it, ask yourself one thing:

> **Who will read this record?**

It needs to be a real person or office. For example:

- a neighbor who is gathering others

- a city or county code officer
- a state environmental agency
- a community contact at the facility
- an expert you plan to hire
- a lawyer
- a reporter

Can't name one? Don't start Section 4 yet. A log nobody reads is one more job for someone who is already tired. Find the reader first. Or keep the log just for your own peace of mind. That is a fine reason too. Just choose it on purpose.

Sending it doesn't guarantee anyone will act. This guide can help you be clear. It can't make anyone answer.

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1. WHAT YOU CAN TRY TONIGHT

Cost: $0. Time: one evening, plus about a week for the bed test (Steps 6 and 7).

Step 1. Rule out your own home. Switch off the fridge, the AC, and then the main breaker for 10 minutes each. Turn each one back on before you switch off the next. If the hum stops, the source is likely inside. Skip this if anyone in your home uses medical equipment, and don't switch off anything unsafe. (Section 4.4 has a longer list of things to check.)

Does the hum seem worse in bed than outside? Your room may be making it stronger.

Every room has deep pitches it boosts. These are called **room resonances**. (The room works like a pipe or a drum.)

Step 2. Measure the room's length, width, and height. Use a tape measure if you have one. If not, step it off heel to toe. One shoe is about one foot.

Step 3. Take one measurement in feet.

Step 4. Divide 563 by that number. The answer is a pitch in **Hz**. (Hz, or hertz, counts pitch. A low number is a deep sound.)

Step 5. Repeat for the other two measurements.

Here is an example for a bedroom about 13 × 11.5 × 8.2 feet:

Measurement	Pitch the room boosts
13 ft	about 43 Hz
11.5 ft	about 49 Hz

Measurement	Pitch the room boosts
8.2 ft	about 69 Hz

Many bedrooms have resonances between **40 and 70 Hz**. Width and height matter too.

Deep sound also tends to be strongest at walls and in corners. Beds are often there.

Step 6. Pick two or three safe places to sleep. Use each one for a few nights.

Step 7. Each morning, write one word: **better**, **worse**, or **same**.

This changes how loud it is where you sleep. It doesn't remove the source.

This might not work. Real rooms are messy. Furniture and doors change things. A spot that calms one pitch can boost another. The middle of the room is not always better.

Feeling worse indoors does not prove the room is the cause. It only means the test is worth a try.

If it helps, good. Count it as one result from a few nights. Not as a promise.

Source note: the room formula is a textbook estimate. It has not been checked against a reference. See Appendix C.

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2. BEFORE YOU SPEND MONEY

Find your hum's pitch before you buy anything. Section 4 shows how. Until then, nobody can tell you what will help. That includes this guide.

A window with two panes of glass has a weak spot. At one pitch, it stops almost nothing. The air between the panes acts like a spring. This is called the **mass-air-mass resonance**. (It is the pitch where glass-air-glass starts to shake along together.)

Here is an example. It uses two 6 mm (about 1/4-inch) panes:

Gap between panes	Weak pitch
2 inches (50 mm)	about 98 Hz
4 inches (100 mm)	**about 69 Hz**
6 inches (150 mm)	about 57 Hz
8 inches (200 mm)	about 49 Hz

With two 1/4-inch panes and a 4-inch gap, the weak spot is near 70 Hz. Thinner glass moves it higher.

This is a rough guide, not an answer.

It only matters if your hum is near the weak pitch. That is usually somewhere from 40 to 80 Hz.

Is your hum somewhere else, like 120 Hz? Then this table tells you nothing about your window. It does not mean the window is fine. It means only real test data can answer.

Real windows also depend on the glass, the frame, the seals, and how it was installed. This table leaves all that out.

Ask the seller:

1. How big is the air gap? How thick is each pane?
2. At what pitch is this exact window weakest?
3. Can I see lab results for each pitch? These are called **transmission loss** data. (They show how much sound the window stops at each pitch.)
4. How much does it stop at **my** pitch? Not the one-number score.
5. Would sealing my old window help as much as new glass?
6. How will we check that it got better?

No pitch-by-pitch lab results? Be careful. A one-number score can hide a weak spot at the exact pitch that keeps you up.

Source note: the window formula is a textbook estimate. It has not been checked against a reference. See Appendix C.

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3. WHY AN OFFICIAL NUMBER CAN READ LOW

Did someone test your noise and say it was low? Here is one possible reason. A low number does not, by itself, mean the test missed your hum. The test may have been taken at a different time or place, and a hum can come and go.

Most noise tests use a filter called **A-weighting**. (A filter is a rule that turns some sounds up or down before they are counted.)

A-weighting copies how ears hear everyday sound. It turns the deep pitches way down.

Here is how much it turns each pitch down. These are filter settings. They are not measurements of your hum.

Pitch	Filter setting
31.5 Hz	−39.4 dB
63 Hz	**−26.2 dB**

Pitch	Filter setting
125 Hz	**−16.1 dB**
250 Hz	−8.6 dB
1000 Hz	0.0 dB

(dB, or decibel, is the unit for sound strength. You only need it to read this table.)

In plain words:

- The standard meter setting counts a 63 Hz hum as about **26 dB quieter** than it is.
- It counts a 125 Hz hum as about **16 dB quieter** than it is.

So a low reading can happen even when a deep hum is real. That alone does not show that a particular test missed your hum.

This is not a trick. A-weighting is the right tool for most noise problems. Most laws are written with it, for good reasons. It was just not made for deep hums.

Some local laws use a different filter. It is called **C-weighting**. (It turns deep sounds down much less.) Some use no filter at all.

Find out which one your area uses. It decides what kind of test could even help your case.

Now you know how to read a report you were given. You also know what to ask for next.

Source note: the filter values come from a published standard (IEC 61672). We have not yet checked the exact edition and table. See Appendix C.

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4. THE SEVEN-NIGHT SCREENING PROTOCOL

What this establishes: frequency, start and stop time, duration, detected recurrence, tonal or broadband character.

What it does not establish: sound pressure level, the source, or whether any rule has been broken.

4.1 The level boundary

Do not report a phone recording as a decibel level. A phone microphone has unknown sensitivity, automatic gain control, and unknown low-frequency response. There is no traceable calibration behind it. An uncalibrated number carried into a meeting is the fastest way to have an otherwise sound record dismissed.

This is a limit on *your phone*, not on affordable measurement generally. A **calibrated Class 2 meter ($200–600)** can report level within its tolerance. If you own one, report level and disclose the device, method, and calibration date. If you don't, report frequency and time — which your phone does far better.

4.2 Recording setup

- Record **uncompressed (WAV)** where available, to avoid extra codec processing. **WAV does not mean raw** — gain control, noise suppression, and equalization may still be applied by the operating system. **[S]**
- Turn off automatic gain control, noise suppression, and voice enhancement wherever the app allows.
- Record at least **60 seconds** per sample.
- Many phone microphones filter out very low sound. If your phone shows nothing but you still hear it, that's not proof it isn't there — note it in your log.
- **Same placement every night.** Write it down: room, height above floor, distance from the nearest wall and window, orientation, case on or off, plugged in or not, microphone unobstructed, HVAC on or off.

4.3 Analysis

- Use a spectrum view with an FFT window of **4 seconds or longer**. A 4 s window gives **0.25 Hz** bins; 10 s gives **0.1 Hz**. **[A]**
- **Real-world accuracy will be worse than bin spacing** — probably around **1 Hz** under good conditions, and this has not been tested. **[E, OPEN]**
- Record the peak frequency, whether the peak is narrow (tonal) or a broad hump (broadband), and any harmonics.
- Free desktop audio software with a spectrum plot is sufficient.

4.4 Rule out your own house first

Before attributing anything outward, check for indoor and nearby sources that cycle: refrigerator, HVAC, ceiling fans, pool pump, aquarium pump, well pump, chargers and power supplies, a neighbor's equipment, road traffic, aircraft.

Do not disable anything essential or unsafe. Just note what is running and, where safe, compare recordings with equipment cycling on and off.

4.5 Wind and weather

Wind on a microphone produces low-frequency noise that can be mistaken for a tone. Log wind and weather every night. Use a foam or fur windscreen outdoors. **Flag and discard** windy nights rather than explaining them — a record that throws out its own inconvenient data is more credible than one that doesn't.

(A wind-speed threshold for reliable outdoor recording has not been established here. TEST PROPOSED — see Appendix D.)

4.6 Indoor and outdoor

Record both on the same night, within a few minutes, using the same placement discipline.

If the tone appears **more prominent indoors across repeated paired observations**, room resonance or structural transmission is worth investigating (Section 1). A single night proves nothing — the source itself may have changed between recordings. Compare the peak's prominence **within each recording**, not levels between recordings.

4.7 Detected recurrence

Report **"detected on 5 of 7 monitored nights."** Never "occurred on 5 of 7 nights."

You may have slept through it, missed a recording, had a phone process the signal away, or had the tone fall below what your microphone captures. **Absence of detection is not confirmed absence.**

Recurrence is still the strongest thing you can produce, because it is the one thing a single professional visit cannot give you. Illustratively: if something occurs on 5 of 7 comparable nights, one randomly chosen night has a **29%** chance of missing it. **[A — illustrative only; consultants may measure multiple nights]**

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5. WHAT YOUR FREQUENCY MIGHT MEAN

Every branch below is **conditional and non-attributing**. A frequency is a lead. It is not an identification.

Near 120 Hz (and 240 / 360 Hz). Electrical transformer cores vibrate at **twice** the supply frequency through magnetostriction — **120 Hz** in the United States, with harmonics at 240 and 360 Hz. A stable tone at these frequencies is **consistent with** electrical equipment, but other equipment also produces them, and frequency alone cannot identify a source. **[S — verify]**

40–80 Hz. The room and window screening in Sections 1 and 2 may be relevant here. Investigate both.

Variable, in the hundreds of hertz. Fans and rotating machinery produce blade-passing tones at RPM/60 × blade count, which shift with load and speed. A tone that wanders may indicate rotating equipment. **[S]**

Broad hump rather than a peak. Broadband noise, not a tone. Different problem, different treatment, and most ordinances handle it better already.

A useful stability test: grid-locked tones hold their frequency almost perfectly. Machinery tones drift with load, temperature, and time of day. Whether your tone stays put across seven nights is informative — and frequency stability is the thing a phone measures best.

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6. WHAT TO DO WITH THE RESULT

Your record says: *"A tone centered near 119 Hz was detected from approximately 12:20 a.m. to 5:00 a.m. on 5 of 7 monitored nights."*

It does not say: what caused it, how loud it was, or that anything is out of compliance.

Language that keeps you credible: *detected*, *consistent with*, *worth investigating*, *screening only*. Language to avoid: *proves*, *caused by*, *in violation*, *measured at X dB*.

Possible next steps: *(Who can act, what to ask each one, and what varies by state: `who-to-send-it-to.md`. A fill-in cover letter: `../log/cover-letter.md`.)*

- **Neighbors** — do others detect the same frequency at the same hours? Multiple independent records are far stronger than one.
- **The facility or operator** — a specific frequency and time window is a far more actionable report than "there's a hum."
- **Local authority** — first ask which weighting their ordinance uses (Section 3). It determines whether their measurement would capture your problem at all.
- **Acoustical consultant** — your seven nights tell them when to measure. That can shorten an expensive engagement considerably.
- **Contractor** — the questions in Section 2, with your measured frequency filled in.
- **Yourself** — the Section 1 experiment, which needs no one's permission.

Keep your evidence reproducible. Store the spectrum images, timestamps, and log. Keep original audio only for events someone may need to independently review, stored locally and encrypted. Tell your household you are recording, avoid placements that capture neighbors' conversations, and delete what you don't need.

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7. THE SEVEN-NIGHT LOG

Printable. One row per night.

#	Date	Start	Stop	Duration	Peak freq.	Tonal / broad	Harmonics	In / Out	Wind & weather	House equipment running	Valid?	Notes
1												

#	Date	Start	Stop	Duration	Peak freq.	Tonal / broad	Harmonics	In / Out	Wind & weather	House equipment running	Valid?	Notes
2												
3												
4												
5												
6												
7												

Placement record (fill once, keep constant): Room ___ · Height ___ · Distance from wall ___ · From window ___ · Orientation ___ · Case on/off ___ · Plugged in? ___ · HVAC ___

One-page summary

> **Low-Frequency Hum Screening Record** Location: _____ Dates monitored: _____ to _____ Nights monitored: ___ Nights with detection: ___ Dominant frequency detected: ___ Hz (range ___ to ___ Hz) Harmonics: _____ Typical start / stop: _____ Longest event: _____ Character: tonal / broadband Nights discarded for wind or other invalidity: ___ Indoor vs outdoor: _____ Household equipment ruled out: _____ **Method:** consumer smartphone, uncompressed recording, FFT analysis, [PHONE / OS / APP]. **Limitations:** no sound-pressure level is claimed; no source is identified; this is a screening record, not a regulatory measurement. Detection counts reflect monitored nights only. Prepared by: _____ Date: _____

APPENDIX A — WHY THIS GUIDE EXISTS

This began as an attempt to convert the noise into electricity. That failed, and the failure produced everything above.

Acoustic intensity: `I = 10^(dB/10) × 10⁻¹² W/m²` **[A]**

Location	SPL	Incident intensity
Residential fenceline	50 dB	0.1 µW/m²
Property line	65 dB	3.2 µW/m²
Louver face	85 dB	316 µW/m²
Fan face	100 dB	10 mW/m²

An impossible ceiling — 20,000 m² of surface at 100 dB, captured at **100%** — yields **200 W** against a hypothetical **1 GW** facility. At a generous 1% conversion, roughly **2 W**: nearly **nine orders of magnitude** below the load. **[A/E]**

> **The hum can be overwhelming to a person and nearly invisible to a power meter.**

Human hearing detects picowatts. **Noise is a perception problem, not an energy problem** — which is precisely why it is so hard to make anyone care about it.

And then the finding that mattered. Five conversion pathways were worked: piezoelectric louvers, a cooling water loop, a resonator field, a distributed resonant wall, machine-mounted harvesting. Every one that closes technically delivers its value to the **operator** — predictive maintenance, event capture, compliance.

The two outputs that help the **resident** — this documentation protocol, and a passive visualizer — produce **no power at all**.

> **The harm cannot be redeemed. It can only be measured, absorbed, or paid for — and every technical escape from that relocates the burden to the party with less standing to object.**

That is why this document is a guide and not a generator.

APPENDIX B — THE VISUALIZER (ARTWORK)

A quarter-wave tube with a tuned mechanical reed at the closed end. Tube length = c/4f: **[A]**

Tone	Length
60 Hz	1.43 m
120 Hz	0.71 m
180 Hz	0.48 m
240 Hz	0.36 m

Four pipes of descending length, played by machinery that never stops.

It is not an instrument. It establishes no level, no source, no duration, no violation. It is a sculpture whose only job is to make a tone visible in a room full of people who have been told it isn't there. Visibility and wind rejection are untested. **OPEN.**

APPENDIX C — WHAT IS NOT VERIFIED

Every claim below is load-bearing and currently uncited or untested. **This appendix is the release condition: do not treat this guide as validated until these are closed.**

#	Claim	Status
1	A-weighting values (IEC 61672 — edition, table, page)	**OPEN** — needs full citation
2	Transformer magnetostriction at 2× supply frequency	**OPEN** — needs authoritative source
3	Mass-air-mass formula, coefficient, and unit convention	**OPEN** — verify against a building-acoustics reference
4	Room-mode formula and its practical limits	**OPEN**
5	Phone frequency accuracy of ~1 Hz	**OPEN** — no configuration tested
6	"WAV does not mean raw"	**OPEN** — platform-specific
7	Wind threshold for valid outdoor recording	**OPEN** — no threshold established
8	Instrument and service price ranges	**[E]** — indicative only
9	Prior art (community noise projects, smartphone calibration research)	**[X]** — three candidate sources identified but unverified
10	A definition of a "qualifying tonal event"	**OPEN** — no detection rule stated

Target: 8–12 verified sources covering items 1–4, 6, and 9. Sources have not been independently checked and may be misattributed or misread.

APPENDIX D — OPEN WORK

This is an Explore-level release. Fancy Ape's obligation here is to define the concept, expose the risks, and organize the next responsible work — not to perform every downstream validation.

Priority	Task	Why	Likely owner	Done when
1	Verify load-bearing citations	Prevents unsupported public claims	Researcher / acoustics student	Claim-to-source bibliography
2	Test phone/app combinations with known tones, including masked and low-level	Establishes which configurations work	Citizen-science group, student, hobbyist	Phone/OS/app results table
3	**Nontechnical usability trial**	Determines whether an ordinary person can follow this at 2 a.m.	One reader, one evening	Observations + revised instructions
4	Define a qualifying tonal event	Prevents subjective event selection	Signal-processing contributor	Published detection rule, e.g. prominence above

Priority	Task	Why	Likely owner	Done when
				adjacent bands
5	Seven-night field trial with a real hum	Establishes real-world usefulness	Resident with recurring hum	Completed log + raw files
6	Compare against a reference meter	Quantifies screening accuracy	Consultant or university lab	Comparison report
7	Evaluate the room-position experiment	Determines whether it repeatably helps	Resident / researcher	Before-and-after across several nights
8	Review window guidance	Prevents bad purchasing advice	Building-acoustics specialist	Frequency-specific contractor worksheet
9	Map real recipients by jurisdiction	Residents need to know who will read this	Community advocate / local government researcher	Contact appendix
10	Establish a wind validity threshold	Prevents false events	Any contributor with an anemometer	Stated threshold + method
11	Test the visualizer	Determines whether the art works	Artist / maker	Prototype video + build notes
12	Publish field revisions	Incorporates evidence without erasing the analytical release	Future maintainer	Controlled revision + change log

Item 3 is the one Fancy Ape should not delegate. Everything else can wait for a contributor. Untested instructions handed to an exhausted person at 2 a.m. is a different kind of failure from an uncited claim, and it costs one evening to close.

Honest position on this handoff: no recipient has been named for this list. Until FancyApe.com's contribution channel exists, the accurate status is *published and open for contribution*, not *handed off*. Section 0 tells a resident not to do seven nights of work without a named reader; the same standard applies to this document.

DOCUMENT NOTES

- **Classification:** Advanced Concept Study — Analytical Screening Guide
- **Method:** DMADV, Level 1 Explore
- **Origin:** Developed from an investigation of data-center noise. Applicable to recurring low-frequency hum from many possible sources.
- **Evidence codes:** P Primary · S Secondary · A Analytical · E Estimate · X Excluded-Unverified
- **Status labels:** OPEN · TEST PROPOSED · VALIDATED *(this document contains no VALIDATED items)*

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ANALYTICAL SCREENING GUIDE — NOT FIELD-VALIDATED · NOT A REGULATORY MEASUREMENT PROCEDURE

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